

Lecture 3: Challenges in Machine Learning

DD2421

Atsuto Maki

Autumn 2026



How should we select/determine the right model f from data?

Basic idea for classification:

Given training data

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$$

of inputs $\mathbf{x}_i \in \mathbb{R}^d$ and their labels y_i .

How should we select/determine the right model f from data?

Basic idea for classification:

Given training data

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$$

of inputs $\mathbf{x}_i \in \mathbb{R}^d$ and their labels y_i .

Compute the misclassification rate on $\mathcal{D}$

$$err(f, \mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \text{Ind}(f(\mathbf{x}_i) \neq y_i)$$

Note: $\text{Ind}(x) = 1$ if $x = \text{TRUE}$ otherwise $\text{Ind}(x) = 0$

- 1 Overfitting
- 2 Cross-Validation
- 3 The Curse of Dimensionality
- 4 The Bias-Variance Trade-off
 - Concept of prediction errors
 - Decomposition of the MSE
 - Bias and variance

Overfitting

Visited in Lecture 2 using decision tree.

Overfitting

Visited in Lecture 2 using decision tree.

Good results on training data, but generalizes poorly.

This occurs due to

- Non-representative sample
- Noisy examples

Overfitting

Visited in Lecture 2 using decision tree.

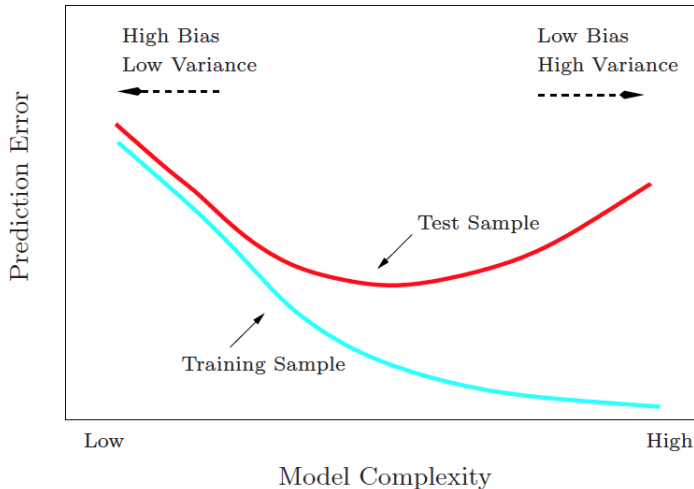
Good results on training data, but generalizes poorly.

This occurs due to

- Non-representative sample
- Noisy examples

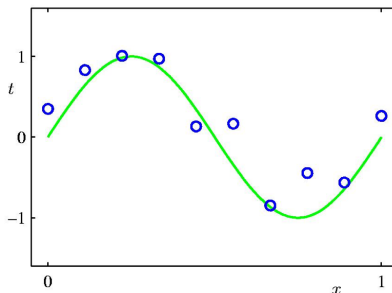
Overfitting

When the learned models are overly specialized for the training samples.



(T. Hastie et al, The Elements of Statistical Learning)

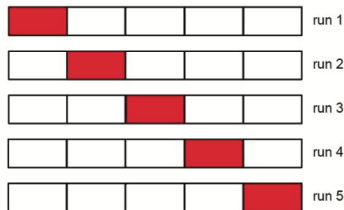
Example: Polynomial Curve Fitting (regression to sinusoidal)



$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

(C. Bishop, Pattern Recognition and Machine Learning)

K -fold cross validation (schematic for $K = 5$)



(K. Murphy, Machine Learning – A probabilistic perspective)

- **Training set** T : to fit the models
- **Validation set** V : to estimate prediction error for model selection (i.e. to determine *hyperparameters*)

If we are in a data-rich situation:

→ partition the data into three sets, *Training set*, *Validation set*, and *Test set* for assessment of the generalization error of the final chosen model.

Curse of Dimensionality

Curse of Dimensionality

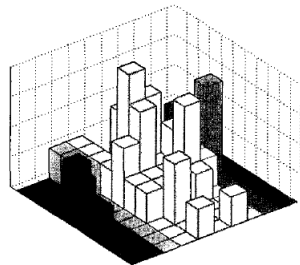
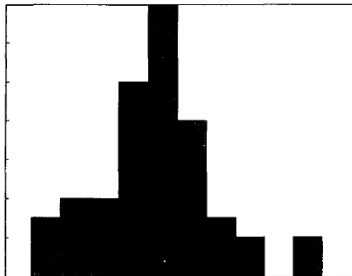
Imagine: inputs represented by 30 features but some of them are less relevant to target function. Will you use all of them?

Curse of Dimensionality

Imagine: inputs represented by 30 features but some of them are less relevant to target function. Will you use all of them?

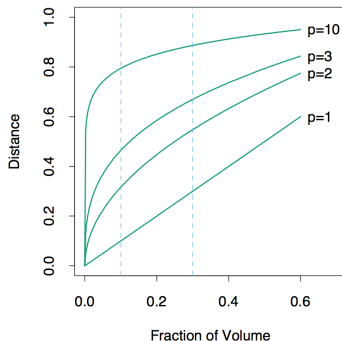
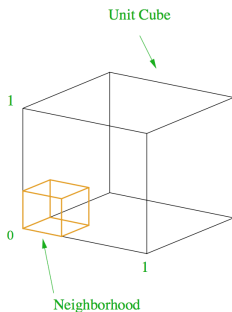
- Easy problems in low-dimensions are harder in high-dimensions
 - training more complex model with limited sample data
- In high-dimensions everything is far from everything else
 - issues in Nearest Neighbours

Example 1: Normal random numbers in 1-d and 2-d
(both plots for 100 inputs)



Too few data to represent the probability density function in 2-d.

Example 2: A subcubical neighbourhood for uniform data in a unit cube.



(T. Hastie et al, The Elements of Statistical Learning)

Graph: The side-length of the subcube needed to capture a fraction of the volume of the data (for different dimensions p).

Intuitions in low-dimensions do not apply in high-dimensions

Intuitions in low-dimensions do not apply in high-dimensions
Real world is in 3-d, but we deal with data for instance in 1000-d

- Uniform distribution on hypercube
- Volume of hypersphere

Intuitions in low-dimensions do not apply in high-dimensions
Real world is in 3-d, but we deal with data for instance in 1000-d

- Uniform distribution on hypercube
- Volume of hypersphere

Techniques for dimensionality reduction / feature selection exist.

The Bias-Variance Trade-off

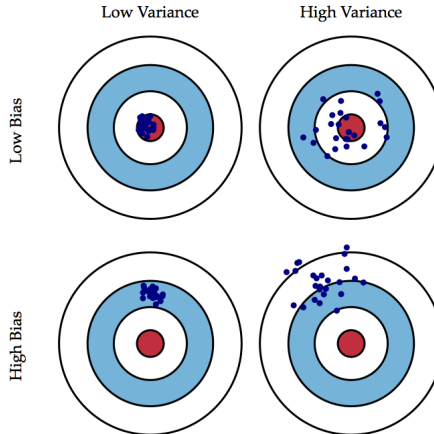
Concepts of prediction errors

Let us imagine we could **repeat** the modeling for many times – each time by gathering new set of training samples, $\mathcal{D}$.

The resulting models will have a **range of predictions** due to randomness in the underlying data set.

- Error due to **Bias**: the difference between the average (expected) prediction of our model and the correct value.
- Error due to **Variance**: the variability of a model prediction for a given data point between different realizations of the model.

Graphical illustration of bias and variance



(figure source: <http://scott.fortmann-roe.com/docs/BiasVariance.html>)

The bias-variance decomposition

Let us consider

$f(\mathbf{x})$: true function

$\hat{f}_{\mathcal{D}}(\mathbf{x})$: prediction function (= model) estimated with $\mathcal{D}$

The bias-variance decomposition

Let us consider

$f(\mathbf{x})$: true function

$\hat{f}_{\mathcal{D}}(\mathbf{x})$: prediction function (= model) estimated with $\mathcal{D}$

and a conceptual tool:

$E_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(\mathbf{x})]$: average of models due to different sample sets
(NOTE: it's shown simply as $E[\hat{f}_{\mathcal{D}}(\mathbf{x})]$ in the sequel)

The bias-variance decomposition

Let us consider

$f(\mathbf{x})$: true function

$\hat{f}_{\mathcal{D}}(\mathbf{x})$: prediction function (= model) estimated with $\mathcal{D}$

and a conceptual tool:

$E_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(\mathbf{x})]$: average of models due to different sample sets
(NOTE: it's shown simply as $E[\hat{f}_{\mathcal{D}}(\mathbf{x})]$ in the sequel)

The mean square error (MSE) for estimating $f(\mathbf{x})$

$$\begin{aligned} E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - f(\mathbf{x}))^2] &= E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])^2] + (E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))^2 \\ &= \mathbf{Variance} + (\mathbf{Bias})^2 \end{aligned}$$

The bias-variance decomposition

Let us consider

$f(\mathbf{x})$: true function

$\hat{f}_{\mathcal{D}}(\mathbf{x})$: prediction function (= model) estimated with $\mathcal{D}$

and a conceptual tool:

$E_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(\mathbf{x})]$: average of models due to different sample sets
(NOTE: it's shown simply as $E[\hat{f}_{\mathcal{D}}(\mathbf{x})]$ in the sequel)

The mean square error (MSE) for estimating $f(\mathbf{x})$

$$\begin{aligned} E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - f(\mathbf{x}))^2] &= E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])^2] + (E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))^2 \\ &= \mathbf{Variance} + (\mathbf{Bias})^2 \end{aligned}$$

To complete, we compute: $E_{\mathbf{x}}[E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - f(\mathbf{x}))^2]]$

The mean square error (MSE) for estimating $f(\mathbf{x})$ is **two-fold**.

The mean square error (MSE) for estimating $f(\mathbf{x})$ is **two-fold**.

$$\begin{aligned}(\hat{f}_{\mathcal{D}}(\mathbf{x}) - f(\mathbf{x}))^2 &= (\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})] + E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))^2 \\&= (\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])^2 + (E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))^2 \\&\quad + 2(\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])(E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))\end{aligned}$$

The mean square error (MSE) for estimating $f(\mathbf{x})$ is **two-fold**.

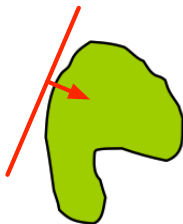
$$\begin{aligned}(\hat{f}_{\mathcal{D}}(\mathbf{x}) - f(\mathbf{x}))^2 &= (\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})] + E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))^2 \\&= (\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])^2 + (E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))^2 \\&\quad + 2(\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])(E[\hat{f}_{\mathcal{D}}(\mathbf{x})] - f(\mathbf{x}))\end{aligned}$$

Taking $E_{\mathcal{D}}[\dots]$ for both sides, the cross term disappears (!) while the second term stays the same.

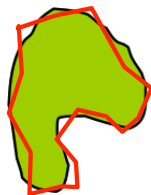
Characterization of a classifier: Bias

Bias of a classifier is the discrepancy between its averaged estimated and true function

$$E[\hat{f}_D(\mathbf{x})] - f(\mathbf{x})$$



High-bias classifier



Low-bias classifier

Low model complexity (small # of d.o.f.) $\implies$ High-bias

High model complexity (large # of d.o.f.) $\implies$ Low-bias

Characterization of a classifier: Variance

Variance of a classifier is the expected divergence of the estimated prediction function from its average value:

$$E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])^2]$$

This measures how dependent the classifier is on the random sampling made in the training set.

Characterization of a classifier: Variance

Variance of a classifier is the expected divergence of the estimated prediction function from its average value:

$$E_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(\mathbf{x}) - E[\hat{f}_{\mathcal{D}}(\mathbf{x})])^2]$$

This measures how dependent the classifier is on the random sampling made in the training set.

Low model complexity (small # of d.o.f.) $\implies$ Low-variance
High model complexity (large # of d.o.f.) $\implies$ High-variance

High variance classifiers produce differing decision boundaries which are highly dependent on the training data.

Also called “flexible”.

Examples:

1. *decision trees*

The depth of the tree determines the variance. How?

2. *k Nearest-Neighbour*

k determines the variance. How?

Our intuition may tell:

- The presence of bias indicates something basically wrong with the model and algorithm...
- Variance is also bad, but a model with high variance could at least predict well on average...

Our intuition may tell:

- The presence of bias indicates something basically wrong with the model and algorithm...
- Variance is also bad, but a model with high variance could at least predict well on average...

So the model should minimize bias even at the expense of variance??

Our intuition may tell:

- The presence of bias indicates something basically wrong with the model and algorithm...
- Variance is also bad, but a model with high variance could at least predict well on average...

So the model should minimize bias even at the expense of variance??

Not really!

Bias and variance are **equally important** as we are always dealing with a single realization of the data set.

Take home message: Match the model complexity to the **data resources**, not to the target complexity